FIBRE PRODUCTS OVER OPEN SUBSCHEMES EMBED AS OPEN SUBSCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3426-3438, following the statement at 3414-3424.

Allowed standard prerequisites (editorial).

- Uniqueness of fibre products. Categories, Section categories-section-fibre-products (cross-chapter, cited at ~3436). Concludes from the exhibited fibre product.

Source reference. Original reference 001U: focused-seeds/geometry/S9-schemes-open-fibre-product/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ and $g : Y \to S$ be morphisms of schemes with the same target. Let $X \times_S Y$, p, q be the fibre product. Suppose that $U \subset S$, $V \subset X$, $W \subset Y$ are open subschemes such that $f(V) \subset U$ and $g(W) \subset U$. Then the canonical morphism $V \times_U W \to X \times_S Y$ is an open immersion which identifies $V \times_U W$ with $p^{-1}(V) \cap q^{-1}(W)$.*

Proof. Let T be a scheme. Suppose $a : T \to V$ and $b : T \to W$ are morphisms such that $f \circ a = g \circ b$ as morphisms into U. Then they agree as morphisms into S. By the universal property of the fibre product we get a unique morphism $T \to X \times_S Y$. Of course this morphism has image contained in the open $p^{-1}(V) \cap q^{-1}(W)$. Thus $p^{-1}(V) \cap q^{-1}(W)$ is a fibre product of V and W over U. The result follows from the uniqueness of fibre products, see Categories, Section 001U. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.